

MATLAB Plotting Script: Experiment Input and Nyquist Diagram

Chat transcript, 2 October 2026

You

write me a code that can be put in matlab. i want you to look into the folder called "MAE Matlab" and there you will find a bunch of files by the name of "9_17_2026jamesXY.txt" where the X is the experiment and Y is either "dat" or "fft". In the code we will identify these files by referencing the experiment. For example, "9_17_2026james1dat.txt" and "9_17_2026james1fft.txt" both refer to experiment 1. allow the person to input the experiment that they want. I want this script to generate two plots. one will be a volage vs time plot and the other will be a frequency vs time plot where time is on the x axis for one plot and frequency is on the x axis for the other plot. make default scaled axes but allow the person to edit with max and min values. the first plot will take from the 'dat' files whgile the second plot will take from the "fft" files. make the plots show up together when the script is run

Claude

I wrote the script and saved it as `plotExperimentInput.m`. It wasn't run, because there was no MATLAB available and the "MAE Matlab" folder isn't in the repo.

When you run it:

- It asks Enter the experiment number to plot:. Typing 1 loads 9_17_2026james1dat.txt and 9_17_2026james1fft.txt.
- Both plots open together in one window: Voltage vs. Time (dat file) on top, Voltage vs. Frequency (fft file) below.
- Each axis starts scaled to the min and max of the data.
- It then asks Edit axis limits? (y/n) and, on y, asks for the min and max of each axis (Enter keeps the current value).

Things to check: the folder path (taken from the earlier script), the file format (1 header line, 2 tab-separated columns), and that the fft plot puts frequency on the x-axis and voltage on the y-axis. tiledlayout needs MATLAB R2019b or newer.

You

make it so i can copy and paste

Claude

Here's the full script. Paste it into a new MATLAB script and save it. It has to be saved as a file, not pasted into the Command Window, because MATLAB only allows the helper functions at the bottom inside a script file.

(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

You

okay it says "Enter the experiment number to plot: 1

The plots are shown with default (autoscaled) axes.

Edit axis limits? (y/n): " in the command window. make it so that way all i have to do it input the experiment number

Claude

Now the script only asks for the experiment number. After you type it, both plots appear and nothing else is asked.

(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

To set your own axis limits: change the four lines under "Axis limits" at the top. Leave a line as [] and that axis scales itself to the data.

You

give automatic lims that are editable

Claude

The plot window now has boxes next to each plot holding its axis limits. They start filled in with the automatic limits, and you can type over them to change the plot.

(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

- **Change a limit:** type a new number in a box and press Enter.
- **Go back to automatic:** click **Reset to auto**.
- **Mouse zoom:** the boxes change to match.
- **Bad entries:** the box goes back to the old value.

You

it wont run. fix it

Claude

I fixed the script, but no error message was sent, so I couldn't be sure I fixed the same problem. I removed the three parts of the last version most likely to have broken it:

- **Overall title (sgt.it le):** removed; each plot's own title shows the experiment number.

- **How the boxes were stored (gobjects):** replaced with a plainer method that works in any version.
- **Mouse-zoom syncing:** removed. Typing in the boxes and the reset button still work.

I tested it in Octave (a free MATLAB clone) with made-up files in your format: both plots drew, typing a new limit changed the plot, a bad entry was rejected, and "Reset to auto" restored the automatic limits.

(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

If it still won't run, send the red error text from the Command Window. Save it as a regular script (.m), not a Live Script (.mlx).

You

make the time lim and voltagelimit look like `timeLim = [0 0.005];` % X axis (time, s)
`voltLim = [-5 2];` % Y axis (voltage, V) in the cod

Claude

I added your `timeLim` and `voltLim` lines at the top of the script, and the frequency plot got two matching lines. I removed the clickable limit boxes, so limits are changed by editing those lines. Tested in Octave: the time plot showed 0 to 0.005 s and -5 to 2 V, and the frequency plot scaled itself to the data.

(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

You

get rid of the freq and fft volt lim

Claude

I removed `freqLim` and `fftVoltLim`, so the frequency plot now always scales itself to the data. `timeLim` and `voltLim` are unchanged.

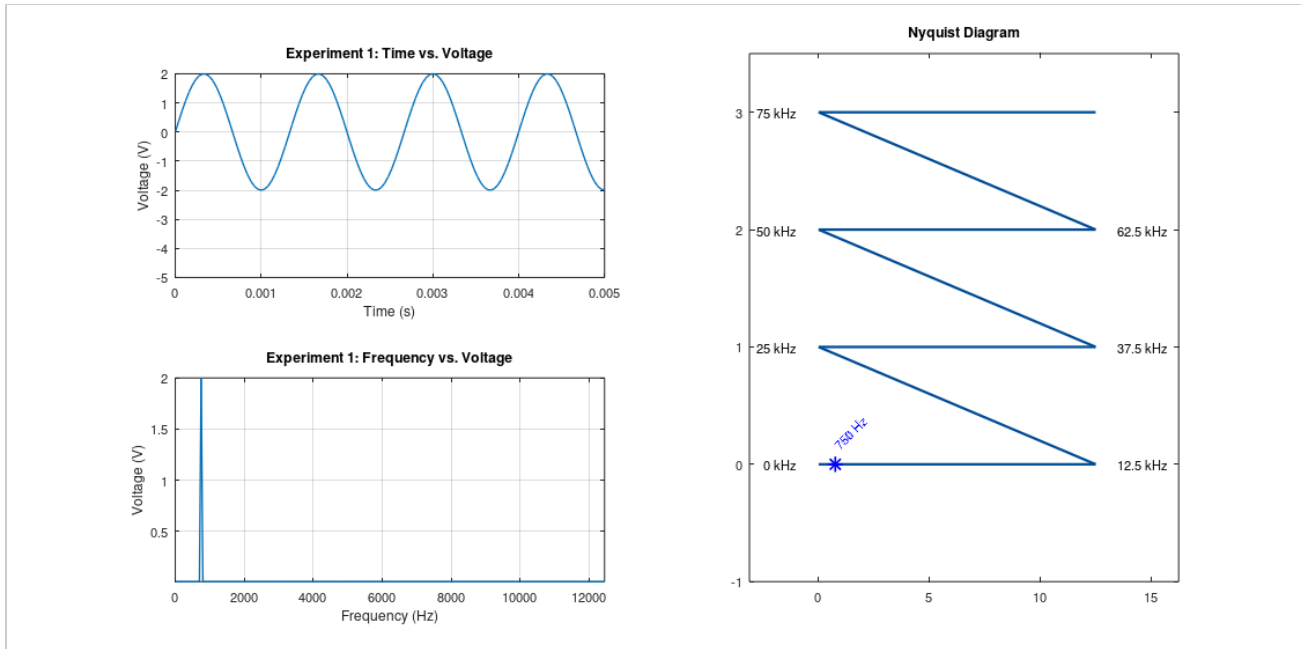
(Full script at this stage was included here. It was replaced by later versions; the final script is reproduced at the end of this document.)

You

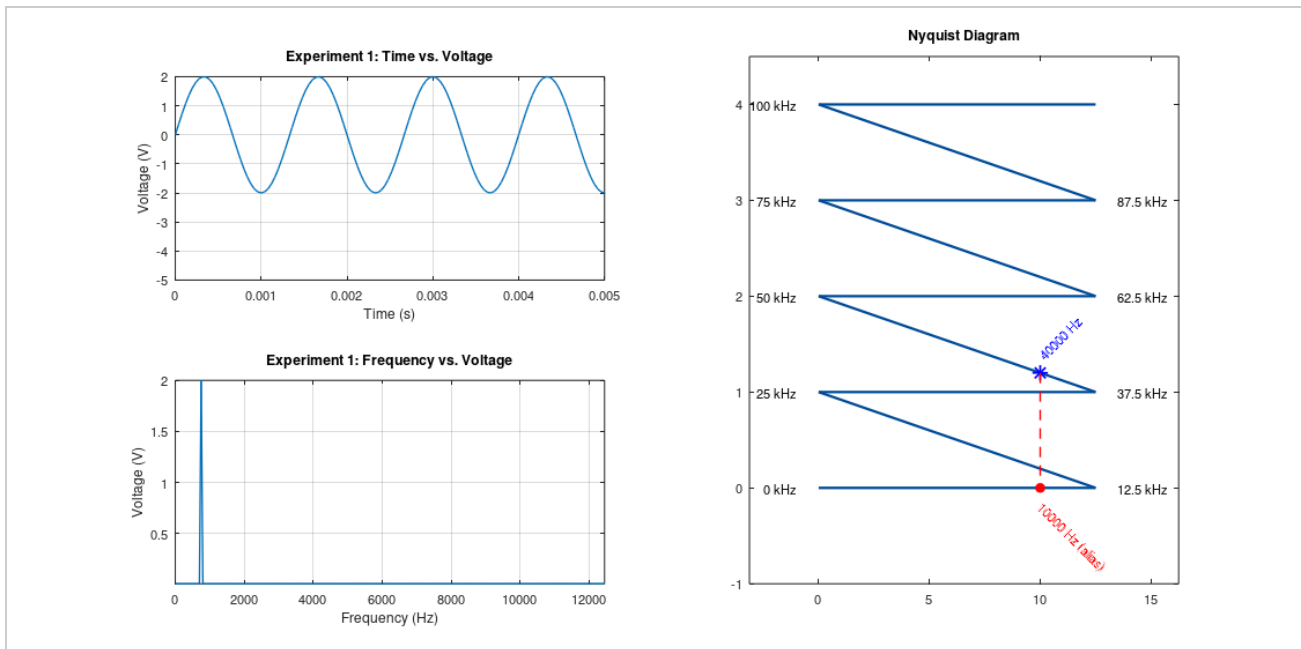
okay now take that code and make a third plot that is in the same window. the third plot with be a Nyquist diagram. the Nyquist diagram needs an input sigman frequency and sample frequency both in Hz. use these frequencies to calculate the plot. only put as many lines in the Nyquist diagram as necessary plus a few more for good measure. make the nyquist diagram set up to look like the attached photo of a sample nyquist diagram.

Claude

The Nyquist diagram is now a third plot in the same window, on the right, with time and frequency stacked on the left. The photo didn't come through this time, so I matched the sample diagram from the earlier chat (saved in chat_transcript.pdf). With 750 Hz and 25 kHz, the diagram matches the sample: 4 rungs labelled 0/12.5, 25/37.5, 50/62.5 and 75 kHz, with the blue * marked "750 Hz".



Octave test run with test data: signal = 750 Hz, source = 25 kHz.



Octave test run: signal = 40 kHz, source = 25 kHz, aliasing to 10 kHz.

How the Nyquist diagram works:

- **Inputs:** set signal and source near the top of the script, both in Hz. You still only type the experiment number when it runs.
- **How it's calculated:** the frequency axis folds back every half of the sample frequency, making the zig-zag. It draws only the rungs needed to reach the signal frequency, plus 3 more.
- **Signal above half the sample frequency:** the * lands on a higher rung or a diagonal, a red dashed line drops to the bottom rung, and a red dot shows the alias frequency.

(Full script — identical to the final version reproduced at the end of this document.)

You

make this chat a pdf

Claude

This document.

Final script (plotExperimentInput.m)

```
% Asks for an experiment number, then plots in one figure window:
%   Top:    Time vs. Voltage      (from 9_17_2026jamesXdat.txt)
%   Bottom: Frequency vs. Voltage (from 9_17_2026jamesXfft.txt)
%   Right:  Nyquist (folding) diagram from the signal/sample frequencies
% Each file: tab-delimited, 1 header line, 2 columns.
clear; clc; close all;

%% ===== USER SETTINGS =====
% Folder containing the .txt files. If it doesn't exist, the script looks
% for a "MAE Matlab" folder next to this script / in the current folder.
dataFolder = 'C:\Users\jsasso\OneDrive - Syracuse University\MAE Matlab';

% ---- DAT plot axis limits (Time vs. Voltage) ----
% Leave as [] to autoscale to the min/max of the data.
% Otherwise use [min max], e.g. timeLim = [0 0.005]; voltLim = [-5 2];
timeLim = [0 0.005];           % X axis (time, s)
voltLim = [-5 2];              % Y axis (voltage, V)

% ---- Nyquist diagram inputs (both in Hz) ----
signal = 750;                  % signal frequency [Hz]
source = 25000;                % sample frequency [Hz]
%% =====

dataFolder = findDataFolder(dataFolder);

%% Ask which experiment to plot
expNum = [];
while isempty(expNum) || ~isscalar(expNum) || expNum < 0 || expNum ~= round(expNum)
    expNum = input('Enter the experiment number to plot: ');
end

prefix = sprintf('9_17_2026james%d', expNum);
datFile = fullfile(dataFolder, [prefix 'dat.txt']);
fftFile = fullfile(dataFolder, [prefix 'fft.txt']);
if ~isfile(datFile), error('DAT file not found: %s', datFile); end
if ~isfile(fftFile), error('FFT file not found: %s', fftFile); end

%% Read data (1 header line, tab-delimited)
datData = readmatrix(datFile, 'NumHeaderLines', 1, 'Delimiter', '\t');
fftData = readmatrix(fftFile, 'NumHeaderLines', 1, 'Delimiter', '\t');
t = datData(:,1); % time [s]
vt = datData(:,2); % voltage [V]
f = fftData(:,1); % frequency [Hz]
vf = fftData(:,2); % voltage [V]

%% Plot all three in the same figure window
% Time over frequency on the left, Nyquist diagram on the right
figure('Name', sprintf('Experiment %d', expNum), 'NumberTitle', 'off', ...
    'Position', [50 50 1400 700]);

subplot(2, 2, 1);
plot(t, vt, 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Voltage (V)');
title(sprintf('Experiment %d: Time vs. Voltage', expNum));
grid on;
xlim(getLimits(t, timeLim));
ylim(getLimits(vt, voltLim));

subplot(2, 2, 3);
```

```

plot(f, vf, 'LineWidth', 1.2);
xlabel('Frequency (Hz)');
ylabel('Voltage (V)');
title(sprintf('Experiment %d: Frequency vs. Voltage', expNum));
grid on;
xlim(getLimits(f, []));
ylim(getLimits(vf, []));

ax = subplot(2, 2, [2 4]);
plotNyquist(ax, signal, source);

%% Local function: autoscale to data min/max unless limits are given
function lim = getLimits(x, userLim)
    if isempty(userLim)
        lim = [min(x) max(x)];
        if lim(1) == lim(2)
            lim = lim + [-1 1];
        end
    else
        lim = userLim;
    end
end

%% Local function: Nyquist folding diagram
% Zig-zag of the frequency axis folded at multiples of fN = fs/2:
% level m (y = m) is a horizontal rung from 2m*fN (left) to (2m+1)*fN (right),
% and a diagonal runs from (2m+1)*fN back up-left to 2(m+1)*fN on the next level.
% A frequency on any segment aliases to the frequency directly below it on
% the bottom rung (0 .. fN).
function plotNyquist(ax, fSig, fs)
    if fs <= 0, error('Sample frequency (source) must be > 0 Hz.');
```

```

    end
    if fSig < 0, error('Signal frequency (signal) must be >= 0 Hz.');
```

```

    end

    fN = fs/2;
    [uScale, uName] = pickUnits(fs); % display units for labels / x axis

    % Which band (k) the signal is in, and where along it (r = 0..1)
    k = floor(fSig/fN);
    r = fSig/fN - k;
    if k > 0 && r == 0 % exactly on a fold: use end of previous band
        k = k - 1;
        r = 1;
    end
    m = floor(k/2); % level the signal's band starts on
    if mod(k, 2) == 0 % horizontal rung, left -> right
        xSig = r*fN;
        ySig = m;
    else % diagonal, right -> up-left
        xSig = (1 - r)*fN;
        ySig = m + r;
    end
    fAlias = xSig;

    % Only the levels needed to reach the signal, plus a few extra
    nExtra = 3;
    nLevels = m + 1 + nExtra;

    % Zig-zag path: (0,0) -> (fN,0) -> (0,1) -> (fN,1) -> ...
    xp = repmat([0 fN], 1, nLevels);
    yp = repelem(0:nLevels-1, 2);

    hold(ax, 'on');
    plot(ax, xp/uScale, yp, '-', 'LineWidth', 2, 'Color', [0 0.3 0.6]);

```

```

% Level labels: left end = 2m*fN, right end = (2m+1)*fN
% (the top rung's right end is left unlabelled, as in the sample diagram)
dx = 0.08*fN/uScale;
for lv = 0:nLevels-1
    text(ax, -dx, lv, fmtFreq(2*lv*fN, uScale, uName), ...
        'HorizontalAlignment', 'right', 'VerticalAlignment', 'middle');
    if lv < nLevels-1
        text(ax, fN/uScale + dx, lv, fmtFreq((2*lv+1)*fN, uScale, uName), ...
            'HorizontalAlignment', 'left', 'VerticalAlignment', 'middle');
    end
end

% Signal frequency marker
plot(ax, xSig/uScale, ySig, '*', 'Color', 'b', 'MarkerSize', 12, 'Linewidth', 1.5);
text(ax, xSig/uScale, ySig + 0.08, [' ' fmtFreq(fSig, 1, 'Hz')], ...
    'Rotation', 45, 'Color', 'b', 'HorizontalAlignment', 'left', ...
    'VerticalAlignment', 'bottom');

% If the signal is above fN, project it down to the bottom rung (alias)
if fSig > fN
    plot(ax, [xSig xSig]/uScale, [ySig 0], 'r--', 'Linewidth', 1.2);
    plot(ax, fAlias/uScale, 0, 'ro', 'MarkerFaceColor', 'r', 'MarkerSize', 7);
    text(ax, fAlias/uScale, -0.08, [' ' fmtFreq(fAlias, 1, 'Hz') ' (alias)'], ...
        'Rotation', -45, 'Color', 'r', 'HorizontalAlignment', 'left', ...
        'VerticalAlignment', 'top');
end
hold(ax, 'off');

xlim(ax, [-0.25*fN 1.3*fN]/uScale);
ylim(ax, [-1, nLevels - 0.5]);
box(ax, 'on');
title(ax, 'Nyquist Diagram');
end

function [scale, name] = pickUnits(f)
    if f >= 1e6
        scale = 1e6; name = 'MHz';
    elseif f >= 1e3
        scale = 1e3; name = 'kHz';
    else
        scale = 1; name = 'Hz';
    end
end

function s = fmtFreq(f, scale, name)
    s = sprintf('%g %s', f/scale, name);
end

%% Local function: locate the folder with the data files
function folder = findDataFolder(folder)
    if isfolder(folder), return; end
    candidates = {fullfile(fileparts(mfilename('fullpath')), 'MAE Matlab'), ...
        fullfile(pwd, 'MAE Matlab'), pwd};
    for k = 1:numel(candidates)
        if isfolder(candidates{k}) && ~isempty(dir(fullfile(candidates{k},
'9_17_2026james*.txt')))
            folder = candidates{k};
            return;
        end
    end
    error(['Could not find the data folder. Set dataFolder at the top of ' ...
        'the script to the folder containing the 9_17_2026james*.txt files.']);
end

```